Lecture 1: Where PDEs Come From

1.1 Suggested Textbooks

- W. Strauss, 1992, *Partial Differential Equations: An Introduction*. John Wiley & Sons.
- R. Haberman, 2004, *Applied Partial Differential Equations*. Prentice-Hall.

1.2 Contents

1. Introductions
   Background materials – general introductions – equations from physics – first order equations
2. Waves and Diffusions
   Wave equation – energy – heat equation – Fourier transformation
3. Boundary problems
   Separation of variables – the Dirichlet condition – the Neumann condition – Fourier series
4. Laplace equation
   Properties – Green function – half-space and sphere

1.3 Background Materials

Here we list some of the standard results from calculous which you should be familiar before starting this section of the course.

1.3.1 Basic facts from calculous

In this subsection, we assume that all the functions are smooth such that they are integrable and infinitely differentiable.

1. Derivatives are *local*. For example, to calculate the derivative $\partial u/\partial x$ at a particular point (x_0, t_0), you just need to know the values of $u(x, t_0)$ for x near x_0, since the derivative is the limit as $x \to x_0$.

2. The chain rule:
$$\frac{\partial}{\partial x}[f(g(x,t))] = f'(g(x,t)) \cdot \frac{\partial g}{\partial x}(x,t). \tag{1.1}$$

3. Derivatives of integrals:
$$\frac{d}{dt}\int_{a(t)}^{b(t)} f(x,t)dx = \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x,t)dx + f(b(t),t)b'(t) - f(a(t),t)a'(t). \tag{1.2}$$

4. Volume integrals: A volume V in $\mathbb{R}^3$ may be described using three parameters, say $\mathbf{x} = (x, y, z) = \mathbf{x}(\xi, \eta, \zeta)$, where (ξ, η, ζ) occupies some region $R \subset \mathbb{R}^3$. Then, given a scalar field $\phi(\mathbf{x})$ and vector field $\mathbf{u}(\mathbf{x})$, we have

$$\iiint_V \phi dxdydz \equiv \iiint_V \phi dV \equiv \iiint_R \phi(\mathbf{x}(\xi,\eta,\zeta))J(\xi,\eta,\zeta)d\xi d\eta d\zeta, \tag{1.3}$$

$$\iiint_V \mathbf{u} dxdydz \equiv \iiint_V \mathbf{u} dV \equiv \iiint_R \mathbf{u}(\mathbf{x}(\xi,\eta,\zeta))J(\xi,\eta,\zeta)d\xi d\eta d\zeta, \tag{1.4}$$

where J is the Jacobian of the transformation from (ξ, η, ζ) to (x, y, z), that is

$$J(\xi,\eta,\zeta) = \frac{\partial(x,y,z)}{\partial(\xi,\eta,\zeta)} := \begin{vmatrix} \partial x/\partial\xi & \partial x/\partial\eta & \partial x/\partial\zeta \\ \partial y/\partial\xi & \partial y/\partial\eta & \partial y/\partial\zeta \\ \partial z/\partial\xi & \partial z/\partial\eta & \partial z/\partial\zeta \end{vmatrix}. \tag{1.5}$$

5. Green Theorem: Let D be a bounded plane domain with a piecewise C^1 boundary curve C. Then

$$\iint_D (q_x - p_y)dxdy = \int_C pdx + qdy. \tag{1.6}$$

6. Divergence Theorem: Let D be a bounded spatial domain with a piece-wise C^1 boundary surface S. Let n be the unit outward normal vector on S. Then

$$\iiint_D \nabla \cdot \mathbf{f} dx = \iint_S \mathbf{f} \cdot n dS, \tag{1.7}$$

where $\nabla \cdot \mathbf{f}$ is the three dimensional divergence of a vector field $\mathbf{f}$.

1.4 General Introductions

A partial differential equation (PDE) is an equation relating an unknown function u of more than two independent variables $u(x, y, z, ...)$ and its partial derivatives. We will often denote its derivatives by subscripts as

$$u_x = \frac{\partial u}{\partial x}, \qquad u_{xy} = \frac{\partial^2 u}{\partial y \partial x}, \tag{1.8}$$

and so forth. We will assume throughout this book, unless stated otherwise, that u is sufficiently smooth for all the required partial derivatives to exist and to be independent of the order in which the derivatives are performed, i.e., $u_{xy} = u_{yx}$. For example, a PDE in two independent variables of first order can be written as

$$F(x, y, u, u_x, u_y) = 0.$$

The *order* of an equation is the highest derivative that appears. The most general second-order PDE in two independent variables is

$$F(x, y, u, u_x, u_y, u_{xx}, u_{xy}, u_{yy}) = 0.$$

A *solution* of a PDE is a function $u(x, y, ...)$ that satisfies the equation at least in some region of the $x, y, ...$ variables.

Some examples of PDEs are:

1. $u_{xx} + u_{yy} = 0$ (Laplace's equation)

2. $u_x^2 + u_y^2 = 0$ (Eikonal equation)

Laplace's equation is *linear*: if $u(x, y)$ and $v(x, y)$ are two solutions, then so is $au(x, y) + bv(x, y)$, where a and b are any two constant scalars. Obviously eikonal equation is *nonlinear*.

Actually PDEs arise in countless real-world situations (such as fluid dynamics, biology, finance,...) and frequently one of the independent variables represents time. In such case, we write $u = u(t, x, y, z, ...)$ instead of $u = u(x, y, z, ...)$, where t is time. For example,

1. $u_t + u_x = 0$ (transport equation)
2. $u_t - u_{xx} = 0$ (heat equation)
3. $u_{tt} - u_{xx} = 0$ (wave equation)

where the *heat equation* describes the flow of heat in one space dimension x and time t, while the *wave equation* governs the propagation of waves along an elastic string.

In general, a PDE does not in general determine u uniquely by itself—boundary conditions (and/or initial conditions) are also required. A PDE supplemented by appropriate boundary conditions (and/or initial conditions) is often referred to as a "problem". An *initial condition* specifies the physical state at a particular time t_0. For *boundary conditions*, there are three important kinds of conditions:

1. u is specified ("Dirichlet condition")
2. the normal derivative $\partial u/\partial n$ is specified ("Neumann condition")
3. $\partial u/\partial n + au$ is sepcified ("Robin condition")

where a is a given function of $t, x, y, z, ...$. A simple problem involving Laplace's equation is

$$\begin{cases} u_{xx} + u_{yy} = 0 & \text{in} \quad y > 0 \\ u = f(x) & \text{along} \quad y = 0 \\ u \to 0 & \text{when} \quad y \to \infty \end{cases} \tag{1.9}$$

Faced with a problem like (1.9), the most important question to ask is: is it *well posed*?

To be *well posed*, a problem must have the following three properties:

1. Existence: a solution $u(x, y)$ exists;
2. Uniqueness: the solution is unique;
3. Stability: the solution depends continuously on the boundary data.

The first of these is obvious: there is no point in trying to find a solution that does not exist. If a problem is physically motivated, and u represents a physical quantity, then we would expect u to have a unique well-defined value at each point. The final condition means that if we vary the function $f(x)$ in (1.9) by a small amount then the corresponding variation in the solution should be similarly small. If it is not, then the solution of the problem is practically impossible, since any numerical errors in $f(x)$, however small, lead to large errors in the solution.

An example of *ill posed* problem is to consider Laplace's equation $u_{xx} + u_{yy} = 0$ in the region $D = \{-\infty < x < \infty, 0 < y < \infty\}$, with specifying both u and u_y on the boundary $y = 0$ (as initial conditions). The reason is that it has the solutions

$$u_n(x, y) = \frac{1}{n} e^{-\sqrt{n}} \sin nx \sinh ny.$$

Notice that they have boundary data $u_n(x, 0) = 0$ and $\partial u_n/\partial y(x, 0) = e^{-\sqrt{n}} \sin nx$, which tends to zero as $n \to \infty$. But for $y \neq 0$ the solution $u_n(x, y)$ do not tend to zero as $n \to \infty$. Thus the stability condition is violated.

Finally, we show how the Laplace, wave and heat equations are in some sense typical among all second-order PDEs. However, these three equations are quite different from each other. In general, let's consider the PDE

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + a_1u_x + a_2u_y + a_0u = 0. \tag{1.10}$$

Theorem 1.1 *By a linear transformation of the independent variables, the equation can be reduced to one of the following three cases:*

1. *Elliptic case: If $a_{12}^2 < a_{11}a_{22}$, then*
$$u_{xx} + u_{yy} + \cdots = 0$$
and we call the PDE is elliptic;

2. *Hyperbolic case: If $a_{12}^2 > a_{11}a_{22}$, then*
$$u_{xx} - u_{yy} + \cdots = 0$$
and we call the PDE is hyperbolic;

3. *Parabolic case: If $a_{12}^2 = a_{11}a_{22}$, then*
$$u_{xx} + \cdots = 0$$
(unless $a_{11} = a_{22} = a_{12} = 0$), and we call the PDE is parabolic.

Here $\cdots$ denotes terms of order 1 and 0.

Proof: For simplicity, assume that $a_{11} = 1$ and always write the lower order term as $\cdots$. Then we can write (1.10) as

$$(\partial_x + a_{12}\partial_y)^2 u + (a_{22} - a_{12}^2)\partial_y^2 u + \cdots = 0 \tag{1.11}$$

where we use the operator notation $\partial_y = \partial/\partial_y$, $\partial_y^2 = \partial^2/\partial y^2$, etc. In the elliptic case $a_{12}^2 < a_{22}$. Let $b = \sqrt{a_{22} - a_{12}^2} > 0$. Introduce the new variables ξ and η by

$$x = \xi, \qquad y = a_{12}\xi + b\eta$$

then $\partial_\xi = 1 \cdot \partial_x + a_{12}\partial_y$, $\partial_\eta = 0 \cdot \partial_x + b\partial_y$, so that the equation becomes

$$\partial_\xi^2 u + \partial_\eta^2 u = 0, \tag{1.12}$$

which is Laplace's. The procedure is similar in the other cases. ■

Exercises

1. Which of the following equations are linear?

$$
\begin{aligned}
&(a) \quad u_x + xu_y = 0 \\
&(b) \quad u_x + uu_y = 0 \\
&(c) \quad u_x + u_y^2 = 0 \\
&(d) \quad u_x + u_y = 0 \\
&(e) \quad \sqrt{1+x^2}(\cos y)u_x + u_{yxy} - [\arctan(x/y)]u = 0
\end{aligned}
$$

2. For each of the following equations, state the order and whether it is nonlinear or not.

$$
\begin{aligned}
&(a) \quad u_t - u_{xx} = 0 \\
&(b) \quad u_t - u_{xx} + xu = 0 \\
&(c) \quad u_t - u_{xxt} + uu_x = 0 \\
&(d) \quad u_{tt} - u_{xx} + u/x = 0 \\
&(e) \quad u_x(1+u_x^2)^{-1/2} + u_y(1+u_y^2)^{-1/2} = 0 \\
&(f) \quad u_x + e^y u_y = 0
\end{aligned}
$$

3. What are the type of the following equations?

$$
\begin{aligned}
&(a) \quad u_{xx} - u_{xy} + 2u_y + u_{yy} - 3u_{yx} + 4u = 0. \\
&(b) \quad 9u_{xx} + 6u_{xy} + u_{yy} + u_x = 0.
\end{aligned}
$$

4. Find the regions in the xy plane where the equation

$$(1+x)u_{xx} + 2xyu_{xy} - y^2 u_{yy} = 0$$

is elliptic, hyperbolic, or parabolic.

1.5 Equations from Physics

In this section we will present a series of examples of PDEs as they occur in physics. They provide the basic motivation for all the PDE problems we study in the rest of the book.

1.5.1 Transport equation

Consider a fluid, water for example, flowing at a constant speed c along a horizontal pipe of fixed cross section in the positive x direction. Let $u(x,t)$ be its density at time t and at position x. Then the amount in the interval $[0,b]$ at the time t is $M = \int_0^b u(x,t)dx$. At the later time $t+h$, the same molecules have moved to the right by ch. Hence

$$M = \int_0^b u(x,t)dx = \int_{ch}^{b+ch} u(x,t+h)dx.$$

Differentiating with respect to b, we get

$$u(b,t) = u(b+ch, t+h).$$

Differentiating with respect to h and putting $h = 0$, we get

$$0 = cu_x(b,t) + u_t(b,t),$$

which is the transport equation

$$\boxed{u_t + cu_x = 0.} \tag{1.13}$$

1.5.2 Wave equation

Consider a flexible, elastic homogenous string or thread, which undergoes relatively small transverse vibrations. For instance, it could be a guitar string. Let $u(x,t)$ be its displacement from equilibrium position at time t and position x. Because the string is perfectly flexible, the tension force is directed tangentially along the string. Let $T(x,t)$ be the magnitude of this tension force. Let ρ be the density (mass per unit length) of the string. It is a constant because the string is homogeneous. Since the motion is purely transverse (only up and down), by Newton's law $F = m\mathbf{a}$, we have

$$\begin{aligned}
&\frac{T}{\sqrt{1+u_x^2}}\Big|_{x_1} - \frac{T}{\sqrt{1+u_x^2}}\Big|_{x_0} = 0, && \text{(horizontal or longitudinal)}\\
&\frac{Tu_x}{\sqrt{1+u_x^2}}\Big|_{x_1} - \frac{Tu_x}{\sqrt{1+u_x^2}}\Big|_{x_0} = \int_{x_0}^{x_1} \rho u_{tt}\,dx, && \text{(vertical or transverse)}
\end{aligned}$$

The left hand sides are the total force on the string, while the right hand sides are the components of the mass times the acceleration integrated over the piece of string.

Because the motion is small, $|u_x|$ is quite small. Then by Taylor expansion,

$$\sqrt{1+u_x^2} = 1 + \frac{1}{2}u_x^2 + \cdots \doteq 1.$$

It means $\sqrt{1+u_x^2}$ my be replaced by 1. Thus from the first equation, T is constant along the string. Let us assume that T is independent of t as well as x. Then take derivatives on the second equation with respect to x_1, we have

$$(Tu_x)_x = \rho u_{tt}.$$

That is

$$\boxed{u_{tt} - c^2 u_{xx} = 0, \qquad \text{where } c = \sqrt{\frac{T}{\rho}}.} \tag{1.14}$$

This is the *wave equation*. Similarly for three dimensional case, the vertical (traverse) component gives (approximately)

$$F = \oint_{\partial D} T\frac{\partial u}{\partial n}ds = \iiint_D \rho u_{tt}\,dxdydz = ma,$$

where the left side is the total force acting on the piece D of the membrane, and where $\partial u/\partial n = n \cdot \nabla u$ is the directional derivative in the outward normal direction, n being the unit outward normal vector on the boundary ∂D. By the divergence theorem, we can deduce the three dimensional wave equation for the simple vibrations as

$$\boxed{u_{tt} - c^2(u_{xx} + u_{yy} + u_{zz}) = 0.} \tag{1.15}$$

Here the operator $\partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is called the *three-dimensional laplacian* operator, usually denoted by $\triangle$.

1.5.3 Heat equation

We let $u(x, y, z)$ be the temperature and let $H(t)$ be the amount of heat contained in a region D. Then

$$H(t) = \iiint_D c\rho u dxdydz,$$

where c is the *specific heat* of the material and ρ is the density. The change in heat is

$$\frac{dH}{dt} = \iiint_D c\rho u_t dxdydz.$$

Fourier's law says that heat flows from hot to cold regions proportionately to the temperature gradient. Thus by the law of conservation of energy, the change of heat energy in D also equals the heat flux across the boundary ∂D,

$$\frac{dH}{dt} = \iint_{\partial D} \kappa(n \cdot \nabla u) dS,$$

where κ is a proportionality factor (*heat conductivity* determined by the Fourier's law). By the divergence theorem,

$$\iiint_D c\rho \frac{\partial u}{\partial t} dxdydz = \iiint_D \nabla \cdot (\kappa \nabla u) dxdydz.$$

If c, ρ, and κ are constants, and let $a^2 = \frac{\kappa}{c\rho}$, then we get the *heat (diffusion) equation*

$$\boxed{u_t = a^2(u_{xx} + u_{yy} + u_{zz}) = a^2 \triangle u.} \tag{1.16}$$

1.5.4 Stationary (steady) waves and diffusions

Consider any of the two previous examples in a situation where the physical state does not change with time. Then $u_t = u_{tt} = 0$. So both the wave and heat equations reduce to

$$\boxed{\triangle u = u_{xx} + u_{yy} + u_{zz} = 0.} \tag{1.17}$$

This is called the *Laplace equation*. Its solutions are called *harmonic functions*.

Exercises

1. Consider the Neumann problem

$$\begin{cases} \triangle u = f(x, y, z) & \text{in } D \\ \frac{\partial u}{\partial n} = 0 & \text{on } \partial D. \end{cases}$$

(a) What can we surely add to any solution to get another solution? So we don't have uniqueness.

(b) Use the divergence theorem and the PDE to show that

$$\iiint_D f(x, y, z) dxdydz = 0$$

is a necessary condition for the Neumann problem to have a solution.

(c) Can you give a physical interpretation of part (a) and/or (b) for heat flow?

1.6 First-Order Linear Equations

We begin our study of PDEs by solving some simple ones. The solution is quite geometric in spirit.

1.6.1 The constant coefficient equation

Let us solve

$$\boxed{au_x + bu_y = 0,} \tag{1.18}$$

where a and b are constants and not both zero.

Geometric method The quantity $au_x + bu_y$ is the directional derivative of u in the direction of the vector $\mathbf{V} = (a, b) = a\mathbf{i} + b\mathbf{j}$. It must always be zero. This means that $u(x, y)$ must be constant in the direction of $\mathbf{V}$. The vector $(b, -a)$ is orthogonal to $\mathbf{V}$. The lines parallel to $\mathbf{V}$ have the equations $bx - ay = constant$. (They are called the *characteristic lines*.) The solution is constant on each such line. Therefore, $u(x, y)$ depends on $bx - ay$ only. Thus the solution is

$$\boxed{u(x, y) = f(bx - ay),} \tag{1.19}$$

where f is any function of one variable.

Characteristic method Let $(x(t), y(t))$ be a line in the (x, y)-plane such that if $u(t) = u(x(t), y(t))$, then the first derivative of $u(t)$ is determined by the transport equation. Such a line $(x(t), y(t))$ is called a characteristic line. By the chain rule,

$$\frac{du(t)}{dt} = \frac{dx(t)}{dt}u_x + \frac{dy(t)}{dt}u_y.$$

Let

$$\frac{dx(t)}{dt} = a, \qquad \frac{dy(t)}{dt} = b.$$

Then the line $(x(t), y(t))$ is

$$x(t) = at + constant, \qquad y(t) = bt + constant,$$

which is $bx - ay = constant$. In addition, along the line,

$$\frac{du(t)}{dt} = au_x + bu_y = 0.$$

Hence $u(x, y)$ is constant along each such line. Thus as the geometric method, the solution is $u(x, y) = f(bx - ay)$, with f an arbitrary function of one variable.

Coordinate method Let's make a change of coordinates to

$$x' = ax + by, \qquad y' = bx - ay. \tag{1.20}$$

Replace all x and y derivatives by x' and y' derivatives. By the chain rule,

$$u_x = \frac{\partial x'}{\partial x}u_{x'} + \frac{\partial y'}{\partial x}u_{y'} = au_{x'} + bu_{y'}$$

and

$$u_y = \frac{\partial y'}{\partial y}u_{y'} + \frac{\partial x'}{\partial y}u_{x'} = bu_{x'} - au_{y'}.$$

Hence $au_x + bu_y = a(au_{x'} + bu_{y'}) + b(bu_{x'} - au_{y'}) = (a^2 + b^2)u_{x'}$. Since $a^2 + b^2 \neq 0$, the equation takes the form $u_{x'} = 0$ in the new variables. Thus the solution is $u = f(y') = f(bx - ay)$, with f an arbitrary function of one variable.

Example 1. Solve the PDE $4u_x - 3u_y = 0$, together with the auxiliary condition that $u(0, y) = y^3$.

By (1.19), we have $u(x, y) = f(-3x - 4y)$. This is the general solution of the PDE. Setting $x = 0$ yields the equation $y^3 = f(-4y)$. Letting $w = -4y$ yields $f(w) = -w^3/64$. Therefore, $u(x, y) = (3x + 4y)^3/64$.

1.6.2 The variable coefficient equation

Characteristic method for general equations For any PDE of the form

$$a(x, y)u_x + b(x, y)u_y = 0.$$

First consider the solution of the ODE

$$\frac{dy}{dx} = \frac{b(x, y)}{a(x, y)}.$$

If the ODE can be solved, so can the PDE, because every solution of the PDE is constant on the solution curves of the ODE. These curves are called the *characteristic curves*.

Example 1. Solve the linear and homogeneous PDE

$$\boxed{u_x + yu_y = 0,} \tag{1.21}$$

with the auxiliary condition $u(0, y) = y^3$.

The ODE

$$\frac{dy}{dx} = \frac{y}{1}$$

has the solutions

$$y = Ce^x.$$

On each of the curves $u(x, y)$ is a constant because

$$\frac{d}{dx}u(x, Ce^x) = \frac{\partial u}{\partial x} + Ce^x\frac{\partial u}{\partial y} = u_x + yu_y = 0.$$

Thus $u(x, Ce^x) = u(0, Ce^0) = u(0, C)$ is independent of x. Putting $y = Ce^x$ and $C = e^{-x}y$, we have

$$u(x, y) = u(0, e^{-x}y).$$

It follows that $u(x, y) = f(e^{-x}y)$ is the general solution of this PDE, where again f is an arbitrary function of only a single variable. Now put $x = 0$, we get $y^3 = f(e^{-0}y)$, so that $f(y) = y^3$. Therefore,

$$u(x, y) = (e^{-x}y)^3 = e^{-3x}y^3.$$

Example 2. Solve the PDE

$$\boxed{u_x + 2xy^2u_y = 0.} \tag{1.22}$$

The characteristic curves satisfy the ODE

$$\frac{dy}{dx} = \frac{2xy^2}{1} = 2xy^2.$$

To solve the ODE, we separate variables: $dy/y^2 = 2xdx$. Hence $-1/y = x^2 - C$, so that

$$y = \frac{1}{C - x^2}. \tag{1.23}$$

Again, $u(x, y)$ is a constant on each such curve. So by solving (1.23) for C, we have

$$\boxed{u(x, y) = f(C) = f(x^2 + \frac{1}{y}),} \tag{1.24}$$

where f is an arbitrary function.

Exercises

1. Solve the first-order equation $2u_t + 3u_x = 0$ with the auxiliary condition $u = \sin x$ when $t = 0$.
2. Solve the equation $3u_y + u_{xy} = 0$. (Hint: Let $v = u_y$.)
3. Solve the linear equation $(1 + x^2)u_x + u_y = 0$.
4. Solve the equation $\sqrt{1 - x^2}u_x + u_y = 0$ with the condition $u(0, y) = y$.